HENRY (YUHAO) ZHOU

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EDUCATION

University of Toronto, Canada — September 2014 - June 2019
Bachelor of Applied Science — Cumulative GPA: 3.92/4.00
Specialist in Computer Engineering — Rank: Top 2%
Minor in Robotics and Mechatronics
Advisor: Prof. Sanja Fidler and Prof. Jimmy Ba

INDUSTRY EXPERIENCE

David AI Labs — December 2025 - Present
Head of Machine Learning — *San Francisco, CA*

- Leading research and production ML across transcripts, text-to-speech, speech-to-speech, evaluation.
- Driving company research presence through technical blog posts and academic publication.
- Leading technical engagements with customers from multiple labs.
- Owning the weekly, quarterly outcomes across research and ML initiatives to support business growth.

Ello Technology, Inc. — March 2023 - November 2025
Chief Research Scientist, Speech — *San Francisco, CA*

- Led the development and deployment of speech perception models powering an AI reading tutor for children.
- Designed the end-to-end ML pipeline, spanning data collection, model architecture, training, and evaluation.
- Shipped models that boosted user satisfaction, driving ARR growth from 1 mil to 5 mil USD in 3 months.

Talka AI — May 2021 - February 2023
Co-founder and AI Architect — *Toronto, Canada*

- Co-founded Talka and drove AI product development, reporting directly to the CTO.
- Built a state-of-the-art speech recognition system for recorded virtual meetings on a $20,000 budget.
- Built an engineering team of five engineers, supported by rotating cohorts of two to four interns.

Facebook, AI Research — August 2019 - August 2020
Facebook AI Resident — *Menlo Park, CA*

- Conducted speech and NLP research with Dr. Michael Auli and Alexei Baevski.
- Co-authored two papers published in NeurIPS and ICASSP.

Nvidia — January 2019 - July 2019
Deep Learning Intern — *Toronto, Canada*

- Conducted computer vision research under the supervision of Prof. Sanja Fidler and Prof. Antonio Torralba.
- Proposed and designed neural network architectures and customized Pac-Man game logic for research experiments.

PUBLICATION

See my full publication record and citations at Google Scholar. 12k+ citations as of 2026-9.

- T. Wu, **Y. Zhou**, W. Ling, H. Yang, J. Veloso, L. Sun, R. Huang, N. Guimaraes, S. Sanner *Towards Dialogue Modeling Beyond Text* (*International Conference on Acoustics, Speech and Signal Processing'23*)
- **H. Zhou**, A. Baevski, M. Auli *A Comparison of Discrete Latent Variable Models for Speech Representation Learning* (*International Conference on Acoustics, Speech and Signal Processing'21*)
- A. Baevski, **H. Zhou**, A. Mohamed, M. Auli *wav2vec 2.0: A framework for self-supervised learning of speech representations* (*Neural Information Processing Systems'20*)
- S. W. Kim, **Y. Zhou**, J. Philion, A. Torralba, S. Fidler. *Learning to Simulate Dynamic Environments with GameGAN* (*Computer Vision and Pattern Recognition'20*)
- **Y. Zhou***, T. Wang*, S. Fidler, J. Ba. *Neural Graph Evolution: Automatic Robot Design* (* denotes equal contribution. *International Conference on Learning Representations'19*)
- **Y. Zhou**, M. Tapaswi, S. Fidler. *Now You Shake Me: Towards Automatic 4D Cinema* (**Spotlight** in *Computer Vision and Pattern Recognition'18*)

SKILLS

Programming	Python, MATLAB, C/C++, JavaScript, Java, SQL, Verilog
Framework	PyTorch, TensorFlow, Flask, NumPy, OpenCV, scikit-learn; GCP, AWS
Markup	HTML/CSS, LaTeX, Markdown

RESEARCH EXPERIENCE

University of Toronto, Vector Institute January 2018 - June 2019
Research Assistant *Advisor: Prof.* ***Sanja Fidler****, Prof.* ***Jimmy Ba***

- Project on robotic structure design using genetic algorithms.
- Proposed using graphs to model RL agent and designed methods to modify graphs for efficient search.
- Designed methods to visualize the evolutionary progress and the genealogy tree in the genetic algorithm.
- Project on using Graph Network for learning dynamics in model-based reinforcement learning.
- Improved graph network for efficiently learning dynamics and discriminating graph isomorphism.
- Open-sourced a customized graph neural network library implemented in PyTorch and TensorFlow.

University of Toronto May 2018 - June 2019
Capstone Research *Advisor: Prof.* ***Stark Draper***

- Project on power-efficient hand gesture classifier for artificial prostheses control.
- Designed and implemented distributed machine learning optimization method using AWS Lambda service.
- Designed gesture classifier on electromyography (EMG) data for real-time processing.
- Benchmarked the performance of deep learning and signal processing methods for gesture recognition.
- Leading the senior-year capstone design team for project management, documentation, and experiments.

University of Toronto January 2017 - November 2017
Research Assistant *Advisor: Prof.* ***Sanja Fidler***

- Project on analyzing physical interaction in movies for automatic annotation of 4D effects in movies.
- Led the project in semantic classification and detection of physical interactions, such as splash, in movies.
- Proposed and designed multimodal neural networks for processing visual and acoustic features. Performed ablation studies on the usefulness of each modality, and optimized the performance of the joint network.
- Created a crowdsourcing website and trained human annotators to get high-quality annotations.
- Researched and trained deep neural networks on various audio and video datasets such as SoundNet.

TALKS AND PRESENTATIONS

Talk Google Brain - Talk on *A Topology Space Odyssey*
Vector Robotics Summer School - Talk on *Self-adaptive Robots and Evolution*
Vector Institute AWS Usage showcase meeting on Neural Graph Evolution

AWARDS

Research John Senders Award (Awarded to 1 Final Design among Engineering Department)
Edith Grace Buchan Scholarship (Summer Research 2018)
University Conference Travel Grant (CVPR18, NeurIPS18, ICLR19)

Academic University of Toronto Graduate with High Honour (2019 Graduates)
Dean's List (2015 - 2019) (Awarded to high academic performing students)

Competition Best First Year Team in UTEK Software Design competition (2015)

ACADEMIC SERVICES

Reviewer ICLR, NeurIPS, ICASSP, IJCAI